

Excitonic Effects on Two-Dimensional Transition-Metal Dichalcogenide Monolayers: Impact on Solar Cell Efficiency

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Cite This: *ACS Appl. Energy Mater.* 2021, 4, 3265–3278



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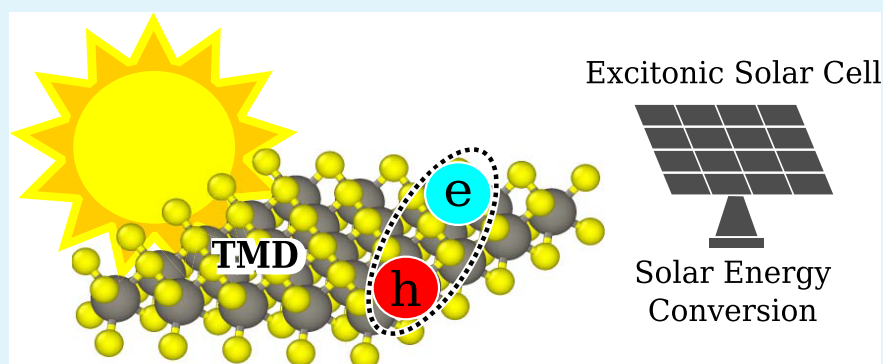
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ABSTRACT: The search for two-dimensional (2D) systems for applications in solar cells has continuously challenged our community. Here, we report screening of 2D monolayers from group IV to XI transition-metal dichalcogenides, MQ_2 , searching for candidates for high-performance photovoltaic devices, where $\text{M} = 3\text{d}, 4\text{d}, \text{ and } 5\text{d}$ transition metals and $\text{Q} = \text{S}, \text{Se}, \text{ and Te}$, i.e., it yields 72 MQ_2 monolayers. Through a robust theoretical framework that combines density functional theory, tight binding based on maximum localized Wannier functions, and the Bethe–Salpeter equation, we investigate the electronic, optical, and excitonic properties of the thermodynamic stable 2H- MQ_2 monolayers. Furthermore, we employ a linear regression analysis of our data to identify nontrivial correlations between different optical and excitonic properties. From the 72 2H- MQ_2 monolayers, we found from phonon calculations that only 22 are dynamically stable in the 2H form, of which 14 are semiconductors with a wide range of energy band gaps. Our results show that the presence of excitons affects the band alignment and the power conversion efficiency. In these compounds, the valley degeneracy breaking driven by spin-orbit coupling and the excitonic correction in the optical properties may be explored, through control of optical helicity, to tune the efficiency of the MQ_2 -based photovoltaic devices. We find high-efficiency van der Waals (vdW) heterostructures for solar cells and observe a strong linear correlation between exciton energy and the energy band gap for the stable semiconductors.

KEYWORDS: 2D materials, transition-metal dichalcogenides, density functional theory, maximum localized Wannier functions, excitons, photovoltaics

1. INTRODUCTION

Since the experimental synthesis of graphene in 2004,¹ the scientific community has been motivated to explore new two-dimensional (2D) materials due to their novel properties.^{2,3} Transition-metal dichalcogenides (TMD) are a major class of 2D graphene cognate, which have gained much attention in virtue of their various atomic configurations and band structures. They consist of hexagonal layers of transition-metal atoms (M) sandwiched between two layers of chalcogen atoms (Q) with an MQ_2 stoichiometry. Depending on the relative alignment of the two Q atoms within the Q–M–Q sandwiched layer, TMDs can have two different polymorph structures: the trigonal prismatic (2H) and the octahedral phases (1T). TMDs can also be found in metallic, semiconductor, and even in superconductor phases, and are highly

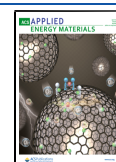
promising for applications in electronic,⁴ spintronic,⁵ valleytronic,⁶ and photovoltaic (PV)⁷ devices. Furthermore, 2D TMDs can be combined to build up van der Waals (vdW) heterostructures⁸ with a wide diversity of properties.

TMD 2H monolayers based on Mo and W have been widely investigated in the literature.^{9,10} They show two inequivalent valleys K and K' degenerated in the first Brillouin zone (BZ), in addition to a non-null Berry curvature.¹¹ The monolayer

Received: December 5, 2020

Accepted: March 31, 2021

Published: April 12, 2021



inversion symmetry breaking, characteristic of the 2H structure, allows the access of valleys degrees of freedom selectively through optical helicity, enabling the probe and the manipulation of charge carriers in both valleys.¹² Furthermore, the strong spin-orbit coupling (SOC) interaction together with the time reversal symmetry leads to a coupling physics of the spin and the valley pseudo-spin degrees of freedom. This is demonstrated not only experimentally—in the selective valley luminescence,¹³ generation of optical valley coherence,¹³ and spectroscopy of magneto-photoluminescence with resolved valley¹⁴—but also theoretically.^{15,16} In this work, we go beyond MoQ₂ and WQ₂ compounds and investigate other 2H-TMD monolayers for application in PV devices.

In a world where 85% of the energy necessities are presently accomplished by fossil fuels and natural gas,¹⁷ the search for renewable alternatives energy sources becomes a need for new generations. Therefore, solar energy has attracted a lot of attention due to its environmental friendliness and low cost. In this scenario, TMD monolayers have been considered reliable candidates for solar cell applications,^{18,19} where the TMD semiconductors with a direct energy band gap possess a high mobility²⁰ and excellent light absorption properties; moreover, their band gap lies in the range of 1–2 eV,^{2,21} matching with solar spectrum.

Although 2D TMDs have been largely studied in the past years, the vast majority of those works are focused on group VI transition metals (TMs), i.e., Mo and W.^{9,22} A complete investigation of other TMD groups that may be combined in efficient heterostructures for PV applications is still missing. Furthermore, owing to the 2D spatial confinement and reduced dielectric screening, TMDs exhibit strong Coulomb interactions that favors the formation of tightly bound excitons upon light excitation. From a theoretical background, the addition of exciton effects may be important to correctly predict the power conversion efficiency (PCE) of TMD-based PV devices.

In this work, we propose a wide investigation of 72 2H-TMD monolayers, composed of the TM from groups IV to XI, period 4–6 of the periodic table, with the chalcogenides S, Se, and Te, proposing vdW heterostructure candidates for high-performance PV devices. We perform a careful study of electronic, optical, and excitonic properties of the thermodynamic stable monolayers. We combine three methods to obtain a detailed knowledge of the PCE of TMD solar cells, namely, density functional theory (DFT), tight binding (TB),^{23,24} and the Bethe–Salpeter equation (BSE).^{10,25} Using our approach for BSE, we have access to the dark excitons (momentum and spin forbidden optical transitions), which do not appear in the absorption spectrum but have relevant effects on the optical properties through nonradiative channels.^{26,27} In general, these states are not described by the usual DFT-GW + BSE approach. Further, using the proposed framework, we have access to the degree of freedom to change the dielectric environment around the monolayers. Therefore, such a theoretical framework is reliable, workable, and requests a lower computational cost, which makes this approach an appropriate choice for this work and future studies.

Throughout our material screening procedure, we observe that only a few systems evince signs of stability in 2H polymorphism, probed via phonon calculations. From the stable systems, a lower number is a semiconductor. Namely, we began with 72 monolayers and found that 14 are stable 2H semiconductors. In the next step, we investigate the excitonic

properties of those 14 monolayers via TB + BSE. Then, we perform a band alignment analysis to discover type II heterojunctions, convenient for TMD solar cells.²⁸ After defining those heterostructures, we estimate the PCE to identify the best TMD heterojunction candidates for PV applications. With the optical and excitonic properties in hand, we performed a complementary search for hidden correlations in our data. In the [Supporting Information](#), we show a detailed roadmap of our calculations ([Section S1](#)) as well as a complete picture of the obtained results.

2. THEORETICAL APPROACH AND COMPUTATIONAL DETAILS

An atomistic investigation of excitonic effects in 2D 2H-TMD monolayers is a challenge,^{29,30} in particular, due to the basic requirements to obtain an accurate description of the electronic properties (energy band gap), SOC effects, and electron-hole Coulomb interactions. In this work, the description of the structural stability, electronic properties, and optical properties, including excitonic effects, was achieved by a sequence of three frameworks, namely, DFT^{31,32} within hybrid exchange-correlation functional,^{33,34} TB^{23,24} using a Wannier90 parametrization,³⁵ and the solution of BSE.³⁶ Furthermore, to obtain insights into and better understanding of the results, we employed linear regression analysis to identify correlations among the data.

2.1. Density Functional Theory Calculations. Our spin-polarized DFT calculations were performed within the generalized gradient approximation (GGA) for the exchange-correlation functional as proposed by Perdew, Burke, and Ernzerhof (PBE).³⁷ To improve the description of the 2H-TMD monolayer lattice parameters, we employed the additive vdW D3 correction proposed by Grimme et al.,³⁸ which contributes to decrease the lattice parameter toward the experimental results.^{39,40} Beyond that, as the DFT-PBE framework tends to underestimate the fundamental energy band gaps due self-interaction errors,^{41,42} the range-separated hybrid functional proposed by Heyd, Scuseria, and Ernzerhof (HSE06)^{33,43} was employed to improve the description of the electronic properties, as well as the structural parameters.

SOC plays a crucial role in the description of the electronic and optical properties of TMD monolayers.^{10,25,44} Further, the lack of lattice inversion symmetry along with the strong SOC in 2H-TMDs results in a coupling of spin and valley pseudo-spin degrees of freedom. Thus, SOC effects were also included in the valence states for both PBE and HSE06 functionals via the second variational approach.⁴⁵ For those SOC calculations, the number of bands was increased to twice the number of electrons to achieve accurate results; see [Section S5](#).

The Kohn–Sham (KS) equations were solved using the projector augmented wave (PAW) method,^{46,47} as implemented in the Vienna *ab initio* simulation package (VASP) version 5.4.4.^{48,49} To obtain the equilibrium lattice constant, a_0 , for the 2H-MQ₂ monolayer, we optimized the stress tensor and atomic forces using a plane-wave cutoff energy of $1.75 \times \text{ENMAX}_{\text{max}}$, where $\text{ENMAX}_{\text{max}}$ is the highest recommended cutoff energy among the M and Q species, while all remaining properties were calculated using a cutoff energy of $1.125 \times \text{ENMAX}_{\text{max}}$. All ENMAX values are summarized within the [Supporting Information \(Table S1\)](#). A total energy convergence criterion of 10^{-6} eV was employed for the KS self-consistent cycle, while the equilibrium structures were

obtained once the atomic forces on every atom were less than 0.010 eV Å⁻¹.

For stress tensor calculations, we employed a *k*-mesh of 16 × 16 × 1 (PBE) and 12 × 12 × 1 (HSE06) for the BZ integrations; however, for the band structure calculations, a *k*-mesh of 16 × 16 × 1 was used for both functional, which was increased toward 32 × 32 × 1 for the density of state calculations. To minimize the spurious interactions between a monolayer and its periodic images along the *z*-direction, a vacuum thickness of 16 Å was set in the monolayer's unit cell.

The harmonic interatomic force constants and phonon dispersions were obtained by the combination of VASP (PBE + D3) and Phonopy package,⁵⁰ using the finite displacement method, a 4 × 4 × 1 2H-MQ₂ supercell, a cutoff energy of 1.125 × ENMAX_{max}, and a *k*-mesh of 3 × 3 × 1. For the phonon calculations, we consider the harmonic approximation,⁵⁰ which has been widely used to investigate the phonon properties of TMDs.⁵¹ A large number of convergence tests are performed, as reported in Section S3.

2.2. Excitonic Effects via Tight-Binding Calculations.

Due to the high computational cost to simulate optical properties with excitonic effect using DFT formalism as implemented in VASP,^{48,49} we resort to BSE through the TB formalism, an accurate and expedite method to calculate excitonic properties.^{10,25} The orthogonal TB Hamiltonian, *H*(*k*), obtained with maximum localized Wannier functions (MLWF) through the Wannier90 code,³⁵ is given by

$$H(\mathbf{k}) = H_0 + \sum_{i=1}^N e^{i\mathbf{k}\cdot\mathbf{R}_i} H_{\mathbf{R}_i} \quad (1)$$

where *H*₀ corresponds to the unitary cell Hamiltonian, which contains the on-site energies and hopping parameters inside the cell. The *H*_{*R*} terms represent the hopping matrices, which describe the interaction between neighbor cells. The matrix elements of *H*₀ and *H*_{*R*} are obtained from the Wannier90 code.³⁵ The number of neighbor cells is directly proportional to the *k*-mesh used for Wannier interpolation. We perform the wannierization for both PBE + D3 and HSE06 + D3 employing the SOC for the valence states. Additional details can be found in Section S6 and Tables S26 and S27.

The excitonic states are calculated by solving the two-particle problem via the BSE equation. The excitonic Hamiltonian, *H*_{exc}, is composed of electron, *H*_e, and hole, *H*_h, single-particle Hamiltonians plus the Coulomb interaction potential, *U*(*R*), which binds the electron-hole pairs, i.e.,

$$H_{\text{exc}} = H_e + H_h + U(\mathbf{R}) \quad (2)$$

where *R* is the separation between the particles. Due to the finite width of 2D systems and the nonhomogeneity of dielectric environment, we adopt a Keldysh-type Coulomb potential^{10,52}

$$U(\mathbf{R}) = -\frac{e^2}{8\epsilon_0\epsilon_d r_0} \left[H_0\left(\frac{|\mathbf{R}|}{r_0}\right) - Y_0\left(\frac{|\mathbf{R}|}{r_0}\right) \right] \quad (3)$$

where *e* is the electron elementary charge and *H*₀ and *Y*₀ are the Struve and Bessel second kind functions, respectively.

In this work, we consider that the 2H-MQ₂ monolayers are supported on the SiO₂ substrate and exposed to the air, which leads to an effective dielectric constant *ε*_d given by the average of air and substrate dielectric constants. Furthermore, *r*₀ corresponds to the characteristic screening length, which

depends on the dielectric constants of the substrate, TMD, and air; a detailed explanation of the procedure to obtain *r*₀ can be found in Section S8. The effect of different dielectric environments on excitonic energies can be seen in Section S9. For periodic systems, it is convenient to work in the reciprocal space, where the Keldysh potential⁵³ acquires the form

$$V(\mathbf{G}) = -\frac{e^2}{2\epsilon_0\epsilon_d|\mathbf{G}|(1 + r_0|\mathbf{G}|)} \quad (4)$$

The excitonic states with center-of-mass momentum *T* can be expanded in terms of a linear combination of the electron and hole wave function product

$$\Psi_{\text{ex}}^n(\mathbf{T}) = \sum_{c,v,\mathbf{k}} [A_{c,v,\mathbf{k},\mathbf{T}}^n (|c, \mathbf{k} + \mathbf{T}\rangle \otimes |v, \mathbf{k}\rangle)] \quad (5)$$

where the indexes *c* and *v* correspond to the conduction band (CB) and valence band (VB) states, with momentum *k* + *T* and *k*, respectively. The exciton eigenvalue equation projected on this basis gives us the BSE,³⁶

$$(E_{c,\mathbf{k}+\mathbf{T}} - E_{v,\mathbf{k}}) A_{c,v,\mathbf{k},\mathbf{T}}^n + \frac{1}{S} \sum_{\mathbf{k}',v',c'} W_{(\mathbf{k},v,c),(\mathbf{k}',v',c'),\mathbf{T}} A_{c',v',\mathbf{k}',\mathbf{T}}^n = E_{\mathbf{T}}^n A_{c,v,\mathbf{k},\mathbf{T}}^n \quad (6)$$

where *E*_{*T*}^{*n*} is the energy of the *n*-th excitonic state with momentum *T* and *A*_{*c,v,k,T*}^{*n*} is the exciton wave function (eigenvector). In eq 6, *E*_{*c,k+T*} − *E*_{*v,k*} corresponds to the energy difference between a CB state *c* with momentum *k* + *T* and a VB state *v* with momentum *k*, and *W*_{(*k,v,c*),(*k',v',c'*),*T*} is the many-body Coulomb interaction matrix element, which can be divided into two parts, direct interaction *W*^d and exchange interaction *W*^x. Furthermore, *S* = *S*_c*N*_k² is the total area of the 2D crystal, *S*_c is the area of crystal unit cell, *a* is the lattice constant, and *N*_k² represents the number of *k*-points in the BZ mesh. In this work, we obtain a good convergence of BSE using *N*_k = 40.^{10,25} Additional details on the BSE framework can be found in Section S7.

To study the optical response of 2H-TMD monolayers and probe the excitonic effects, we calculate the absorption spectrum without and with excitons. For the former case, the transition rate of direct optical transitions can be given by Fermi's golden rule^{10,15}

$$I(\mathbf{P}, \omega) = \frac{\pi}{\omega S} \sum_{\mathbf{k},c,v} |\langle c, \mathbf{k} | H_{\text{LM}}(\mathbf{P}, \mathbf{k}) | v, \mathbf{k} \rangle|^2 \delta(\hbar\omega - (E_{c,\mathbf{k}} - E_{v,\mathbf{k}})) \quad (7)$$

where *H*_{LM}(*P*,*k*) represents the light–matter interaction operator (see Section S7). *P* corresponds to the light polarization, *ω* is the absorbed photon frequency, *c*(*v*) labels the conduction(valence) band, and *E*_{*c*(*v*),*k*} is the TB eigenvalues of conduction (valence) state. We showed the absorption spectrum as an alternative of absorption coefficient, as the underlying physics are the same. The formalism used to obtain the spectra is similar to the expression used to obtain the imaginary part of the dielectric function.

The optical activity in the entire BZ can be calculated using the following expression

$$I_{\text{BZ}}(\mathbf{P}, \mathbf{k}) = \frac{\pi}{S} \sum_{c,v} \frac{|\langle c, \mathbf{k} | H_{\text{LM}}(\mathbf{P}, \mathbf{k}) | v, \mathbf{k} \rangle|^2}{(E_{c,\mathbf{k}} - E_{v,\mathbf{k}})} \quad (8)$$

The direct exciton ($\mathbf{T} = \mathbf{0}$) absorption intensity is also obtained via Fermi's golden rule^{10,25}

$$I_{\text{exc}}(\mathbf{P}, \omega) = \frac{\pi}{\omega S} \sum_M O_s(\mathbf{P}, M) \delta(\hbar\omega - E_0^M) \quad (9)$$

where E_0^M is the M th direct excitonic state energy and $O_s(\mathbf{P}, M)$ is the oscillator strength, given by

$$O_s(\mathbf{P}, M) = \left| \sum_{\mathbf{k}, c, v} A_{c,v,\mathbf{k},0}^M \langle c, \mathbf{k} | H_{\text{LM}}(\mathbf{P}, \mathbf{k}) | v, \mathbf{k} \rangle \right|^2 \quad (10)$$

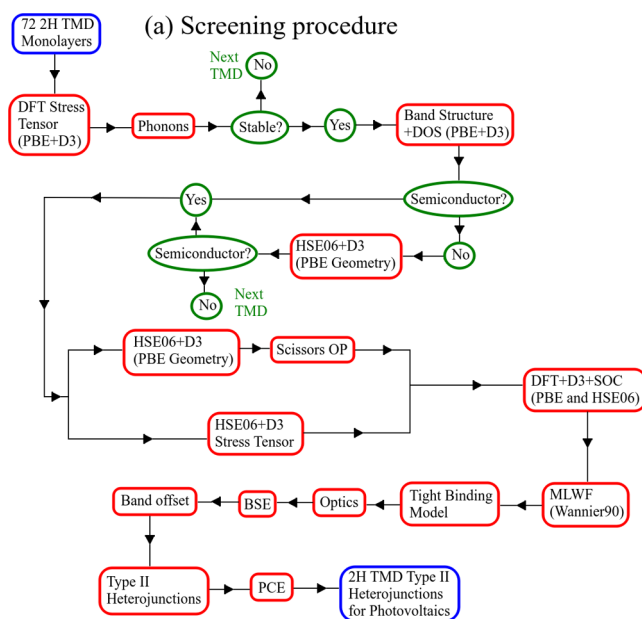
Optical absorption including indirect excitons ($\mathbf{T} \neq \mathbf{0}$) can be driven by phonons in systems at high temperatures and are beyond the scope of this work. To finish our investigation of excitonic properties, in Section 5, we show the relation with exciton energy and monolayer direct energy band gap together with the effect of different dielectric environment.

2.3. Two-Dimensional Transition-Metal Dichalcogenides. Several structures have been reported for the TMDs, namely, 2H, 1T, and 3R; however, the 2H hexagonal structure with space group D_{3h} yields the lowest energy configuration for some systems,^{54,55} and a rich valley physics; hence, it will be employed in this work. For M, we selected TMs from group IV (Ti, Zr, Hf), group V (V, Nb, Ta), group VI (Cr, Mo, W), group VII (Mn, Tc, Re), group VIII (Fe, Ru, Os), group IX (Co, Rh, Ir), group X (Ni, Pd, Pt), and group XI (Cu, Ag, Au), which yields a total of 72 possible 2H-MQ₂ monolayers, as shown in Figure 1(b).

Several of those compounds might be metallic or even unstable, which requires a screening analysis before we address the excitonic effects. Figure 1(a) displays a representation of our screening sequence to select the most suitable 2D-TMD monolayers for PV applications, which are composed of the following steps: (i) initially, we performed a DFT-PBE + D3 stress tensor calculation to obtain an optimized crystal structure for all 72 compounds using a (1×1) unit cell. (ii) To check the stability, we performed phonon calculations using VASP + Phonopy. The MQ₂ monolayers without imaginary frequencies are considered stable^{51,56} and analyzed in the next steps. An example of a stable and an unstable system can be seen in Figure 2(a) and (b), respectively. (iii) Among the stable systems, we calculated the band structure and density of states using both PBE + D3 and HSE06 + D3 functionals, which were employed to separate the systems in metallic and semiconductors. See Section S4 for detailed results of the screening procedure.

Once the semiconductor 2H-MQ₂ systems were identified, we performed the following steps: (iv) HSE06 + D3 calculations using the PBE + D3 lattice parameters to obtain the scissors (SCS) operator to be used in TB parameterization of PBE + D3 + SOC (see Table S21 for those values). (v) HSE06 + D3 stress tensor calculation. (vi) PBE + D3 + SOC and HSE06 + D3 + SOC calculations to compare the SOC effects on both functionals. (vii) Thus, the wannierization is performed using the PBE + D3 + SOC + SCS and HSE06 + D3 + SOC band structures to obtain the TB Hamiltonians, which will be used to study the excitonic effects.

2.4. Correlation Analysis. In this work, we perform a large number of calculations and obtain several data, which are



(b) Transition-metal dichalcogenides - MQ₂

4	5	6	7	8	9	10	11	
22	23	24	25	26	27	28	29	16
Ti	V	Cr	Mn	Fe	Co	Ni	Cu	S
40	41	42	43	44	45	46	47	34
Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Se
72	73	74	75	76	77	78	79	52
Hf	Ta	W	Re	Os	Ir	Pt	Au	Te

Figure 1. (a) Workflow for screening procedure. (b) Transition-metal dichalcogenides MQ₂, M = 3d, 4d, 5d transition metals and Q = S, Se, and Te, and it yields 72 TMD compounds.

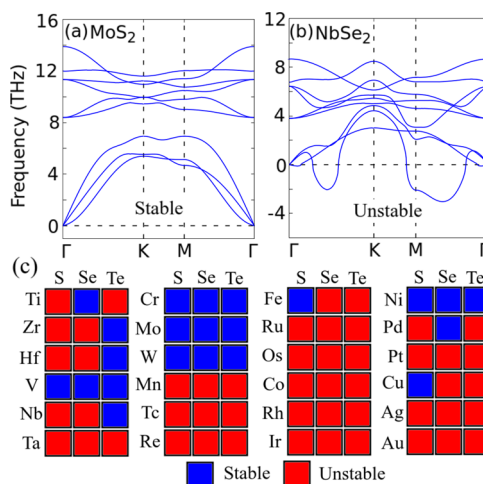


Figure 2. Phonon dispersion of (a) a stable and (b) an unstable 2H-TMD monolayer. (c) Blue (red) color indicates the list of the 22 stable (50 unstable) 2H-TMD monolayers.

discussed based on physical principles. To search for additional correlations in our data, we use it to fit a model that assumes the functional shape

$$F_i = \sum_j (w_j x_j) + b_i \quad (11)$$

where F_i is the i th numerical property we aim to model, x_j is the j -th input properties we are choosing as dependencies of F_i ,

while w_{ij} is the weight associated with the j -th variable in the modeling of the i th property and b_i is the bias (or independent term) in model of the i th property. From a general point of view, this model is equivalent to a single-node artificial neural network using identity as the activation function. So, the training (or fitting) of this model was done with Ordinary Least Squares from `sklearn` package.⁵⁷

We use the energy band gaps, the bright exciton ground-state energy, and the SOC degeneracy breaking of the stable semiconductors as the F property; the x variables (inputs) are the lattice constant, the M–Q distance, the Q–M–Q angle, the Q–Q distance, the period of M, the group of M, and the period of Q. The original x data are also processed using `sklearn`'s `StandardScaler` preprocessing function, which does not change the accuracy of the model (i.e., `StandardScaler` does not influence the value of R^2). With this preprocessing, the weights w_{ij} do not depend on units and give a quantitative value for the strength of the correlation between F_i and x_j .

3. RESULTS AND DISCUSSION

3.1. Phonon Stability Analysis. From the initial 72 2H-MQ₂ monolayers, we found 22 stable systems, which are indicated as blue squares in Figure 2(c). Figure 2(a) and (b) shows the phonon spectrum of stable and unstable monolayers, respectively. The presence of imaginary frequencies, represented as negative values in the phonon dispersion, indicates unstable monolayer. This may be understood through a classical phonon analysis (Newton's equations of motion for the harmonic oscillations): an imaginary frequency leads to non-oscillatory motion. The ion does not remain around an equilibrium position but rather moves away, which can be interpreted as instability. The dispersion of the 72 2H-MQ₂ monolayers is shown in Section S4.2.

We observe that all of the 2H-TMD monolayers of groups VII and IX are unstable, while group VI presents only stable systems. For the other groups, we found stability in the following systems, group by group. Group IV: TiSe₂, ZrTe₂, and HfTe₂; group V: VS₂, VSe₂, VT₂, and NbTe₂; group VIII: FeS₂; group X: NiS₂, NiSe₂, NiTe₂, and PdSe₂; group XI: CuS₂. In the stable systems, as the chalcogenide mass decreases, from Te to S, the M–Q bonds become more ionic; consequently, the width of acoustical modes decreases and the highest frequency of optical modes also increases. This was reported in previous works for 2H and 1T phases.⁵¹ Longitudinal acoustic (LA) modes have a linear dispersion as $k \rightarrow 0$, whereas out-of-plane (ZA) acoustic modes display parabolic dispersion owing to the rapid decay of transverse force constants.

The phonon dispersion of acoustic modes in CrSe₂, CrTe₂, MoTe₂, WTe₂, NiTe₂, and PdSe₂ exhibits a local minima in the BZ (see Figures S20 and S24), which can be associated with Kohn anomalies, and leads to discontinuities in phonon velocities.⁵⁸ This happens in a specific k -vector in BZ, where the dielectric constant features a singularity, causing an abrupt change in the electron screening of the lattice potential.⁵¹ The type of symmetry of the stable monolayer strongly depends on the filling of the d -orbitals of the transition metals. For the 2H phase, the d -states splits into three degenerate states d_z^2 , $d_{x^2-y^2}$, and $d_{xy,yz}$ states.⁵⁹ The group VI (d^2) and most of group IV (d^0), for instance, have trigonal prismatic phases. Group V (d^1) TMDs could exhibit both trigonal prismatic and octahedral structures, whereas group VII (d^3) TMDs generally show distorted octahedral structures, being unstable in trigonal prismatic phase.

3.2. Metallic and Semiconductor Monolayers. Among the 22 2H-TMD stable monolayers, we found that 7 systems show metallic behavior, while 15 compounds—from groups IV, V, and VI—are semiconductors, as shown in Figure 3(e)

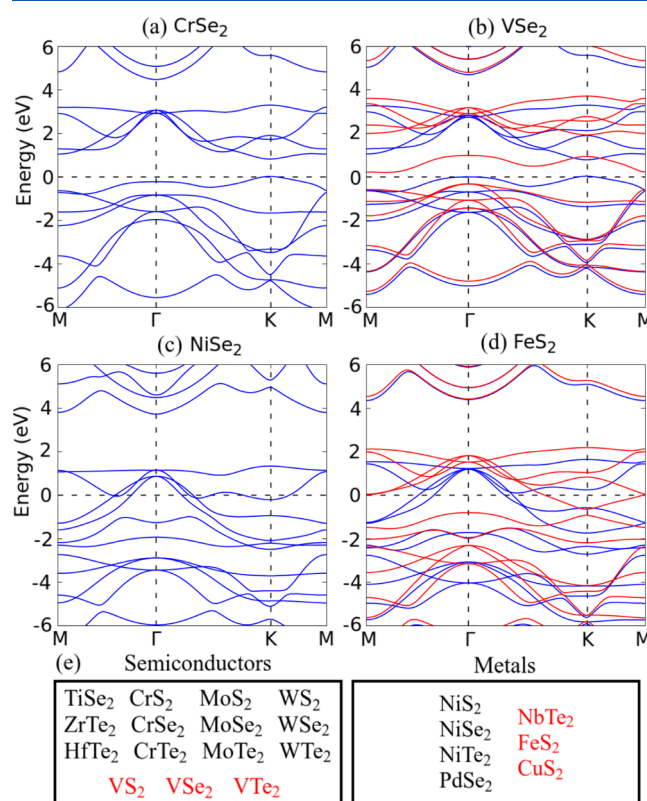


Figure 3. DFT-PBE + D3 band structure of stable 2H-TMD (a) semiconductor (CrSe₂), (b) magnetic semiconductor (VSe₂), (c) metallic (NiSe₂), and (d) magnetic metallic (FeS₂) monolayers. In the magnetic systems, the blue (red) curves represents spin-up (down) states. The Fermi level is set at 0 eV. (e) List of stable semiconductors and metals; the systems in red are magnetic.

(magnetic materials are highlighted in red). Figure 3 (a–d) displays examples of the PBE + D3 band structure of different classes of materials. The band structure of CrSe₂ is shown in Figure 3(a) as an example of nonmagnetic semiconductor, with a direct band gap of 0.84 eV (PBE + D3) at K point. VSe₂ (Figure 3(b)) is an example of magnetic semiconductor, with nondegenerate spin sub-bands. NiSe₂ (Figure 3(c)) and FeS₂ (Figure 3(d)) are nonmagnetic and magnetic metals, respectively.

The p -states of the chalcogen atoms are located at a much lower energy than the Fermi level; therefore, the filling of the d -states drives the nature of the 2H-TMD compounds. In general, completely filled orbitals give rise to semiconducting materials, while partial filling leads to metallic behavior. The stable monolayers from groups IV and VI have an even number of valence electrons in the unit cell (10 and 6, respectively); therefore, they are likely to feature a finite energy band gap.⁶⁰ Group V materials, on the other hand, have an odd number of electrons in their orbitals, being metallic in 1T form.⁶¹ Our results reveal, however, that they are magnetic semiconductors in the 2H structure.

Although the PBE + D3 calculations yield reliable results for the screening procedure, it underestimates the energy band

Table 1. Structural and Electronic Properties of the 2H Stable Semiconductor TMD Monolayers: Lattice Parameter, a_0 , Direct (indirect) Energy Band Gap, $E_g^{\text{dir(ind)}}$, Direct (Indirect) Energy Band Gap with SOC, $E_{g,\text{SOC}}^{\text{dir(ind)}}$, Spin-Orbit Degeneracy Breaking in VBM between K and K' Valleys for a State with Same Spin, $\delta_{k,k'}^{\text{SOC}}$

System	XC	a_0 (Å)	E_g^{dir} (eV)	E_g^{ind} (eV)	$E_{g,\text{SOC}}^{\text{dir}}$ (eV)	$E_{g,\text{SOC}}^{\text{ind}}$ (eV)	$\delta_{k,k'}^{\text{SOC}}$ (meV)
TiSe ₂	PBE	3.47	0.89	0.57	0.84	0.51	3.74
	HSE06	3.46	1.82	1.44	1.73	1.37	5.74
ZrTe ₂	PBE	3.91	0.64	0.42	0.54	0.27	10.60
	HSE06	3.90	1.28	1.05	1.14	0.84	10.45
HfTe ₂	PBE	3.87	0.68	0.27	0.58	0.08	39.80
	HSE06	3.75	0.93	0.46	0.88	0.23	43.44
VS ₂	PBE	3.16	metal		metal		30.92
	HSE06	3.14	1.03	dir gap	1.00	dir gap	64.19
VSe ₂	PBE	3.32	0.76	0.22	0.70	0.16	47.11
	HSE06	3.29	0.99	dir gap	0.91	dir gap	123.50
VTe ₂	PBE	3.55	0.41	0.16	0.37	0.07	49.23
	HSE06	3.54	1.18	0.71	1.11	0.51	31.19
CrS ₂	PBE	3.02	1.04	dir gap	0.94	dir gap	69.24
	HSE06	2.97	1.49	dir gap	1.41	dir gap	132.52
CrSe ₂	PBE	3.18	0.84	dir gap	0.74	dir gap	92.14
	HSE06	3.12	1.29	dir gap	1.14	dir gap	237.59
CrTe ₂	PBE	3.48	0.57	dir gap	0.45	dir gap	108.42
	HSE06	3.38	1.10	dir gap	metal		0.00
MoS ₂	PBE	3.16	1.84	dir gap	1.66	dir gap	148.62
	HSE06	3.13	2.30	dir gap	2.19	dir gap	192.67
MoSe ₂	PBE	3.29	1.60	dir gap	1.41	dir gap	185.40
	HSE06	3.25	2.07	dir gap	1.90	dir gap	258.18
MoTe ₂	PBE	3.51	1.22	dir gap	1.02	dir gap	217.17
	HSE06	3.47	1.67	dir gap	1.44	dir gap	336.93
WS ₂	PBE	3.17	2.00	dir gap	1.60	dir gap	420.53
	HSE06	3.13	2.44	dir gap	2.14	dir gap	507.53
WSe ₂	PBE	3.29	1.76	1.71	1.36	1.28	451.99
	HSE06	3.25	2.20	dir gap	1.86	1.81	565.73
WTe ₂	PBE	3.51	1.28	dir gap	0.86	dir gap	473.01
	HSE06	3.47	1.67	dir gap	1.25	dir gap	626.59

^aThe Grimme D3 van der Waals correction was added for all calculations. The systems in bold are magnetic.

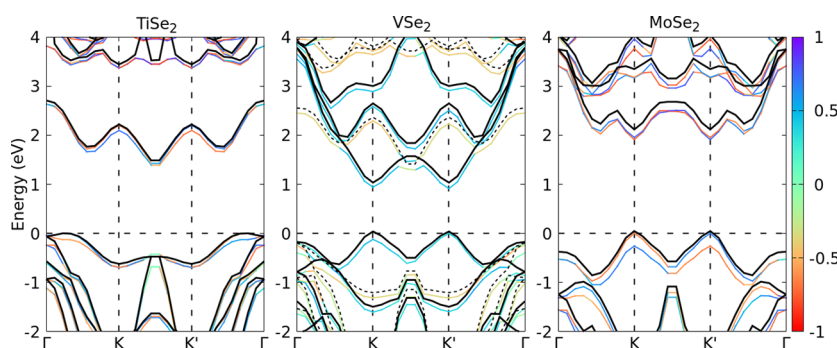


Figure 4. DFT-HSE06 + D3 + SOC band structure of stable semiconductor 2H-TMD monolayers of group IV (TiSe₂), group V (VSe₂), and group VI (MoSe₂). The color code represents the average value of S_z operator. The black curves correspond to the calculation without SOC; in VSe₂, the solid black curves correspond to spin-up states and the dashed ones correspond to spin-down states in the spin-polarized calculation.

gap.^{9,55} Therefore, the electronic and optical properties of the 15 2H-TMD semiconductors are further investigated using the hybrid DFT-HSE06 + D3 framework. Our results obtained with both functionals are summarized in Table 1.

For the same transition metal, the lattice constant, a_0 , increases for heavier chalcogens with both functionals. Furthermore, for fixed Q (S, Se, or Te), a_0 decreases as we move from group IV to group VI. Comparing the PBE + D3 and HSE06 + D3 results, we observe that a_0 is slightly larger with PBE + D3. Concerning the energy band gaps, we also

observe the opposite trend with PBE + D3 and HSE06 + D3 calculations: the lower values are found for V-based material. Moreover, the energy band gap is direct for group VI; with the exception of WSe₂, however, the difference between direct and indirect band gaps is only 5 meV for this monolayer. Group IV materials, on the other hand, show indirect band gaps, where the difference between direct and indirect gaps is of the order of tens of meV, which have important effects on the optical properties and exciton spectra, as described in Section 3.5.

As reported in the literature,^{9,55} in general, PBE + D3 underestimates the energy band gap. Important qualitative differences between the functionals are observed in group V. VS_2 is metallic with PBE + D3, while the gap opens with HSE06 + D3. Furthermore, VSe_2 goes from indirect to direct band gap as we improve the electronic calculation. The four rightmost columns of Table 1 display SOC effects, which will be explored in the next subsection.

3.3. Role of SOC Effects in the Electronic Structure.

SOC plays a crucial role in the electronic structure of stable semiconductors; it tunes the direct and indirect energy band gaps as shown in Table 1. From the PBE + D3 and HSE06 + D3 results, we observe that the former functional underestimates the SOC effects. For both functions, it is larger in K point, ranging from a few to hundreds of meV. It increases from group VI to group IV, and tungsten-based materials show the largest effects, due to the heavier transition metal. The spin-selective VB splitting have important consequences on the excitonic effect, leading to two well-separated absorption peaks, as described in Section 3.5.

Figure 4 shows an example of the HSE06 + D3 band structure for each group, without (black lines) and with (color lines) SOC—the color code indicates the spin mean value, $\langle S_z \rangle$. In the left panel, TiSe_2 is a representative example of group IV TMD monolayer (the corresponding results for ZrTe_2 and HfTe_2 can be seen in Figure S43a(a)). It features an indirect Γ – M energy band gap of 1.37 eV (1.44 eV) with (without) SOC, while the first optical transition is observed at M point (see Figure 5(a)). Due to the indirect energy band gap, the optical properties are slightly affected by SOC.

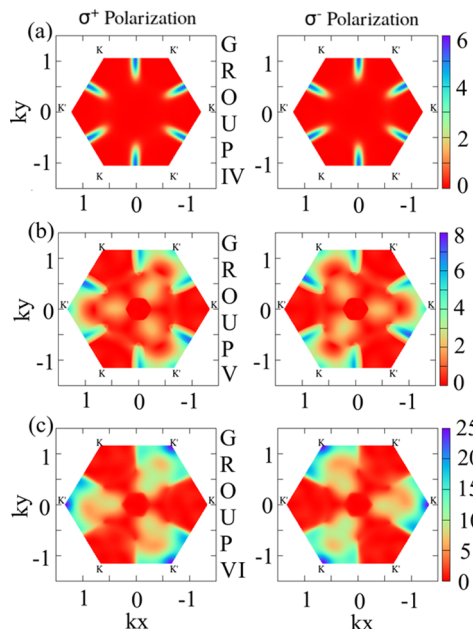


Figure 5. Optical activity in BZ through optical helicity for 2H-TMD monolayer from (a) group IV (TiSe_2), (b) group V (VS_2), and (c) group VI (MoS_2).

Figure 4 (middle) shows an example of V-based electronic structure, the VSe_2 (the other vanadium-based materials show similar results, see Figure S43(b)). The intrinsic magnetic moment breaks the spin degeneracy, as shown by solid and dashed black lines without SOC. The Fermi level is located exactly between the spin-split VBs, with a half-filled K/K'

valley. More importantly, we observe that the bands at the two nonequivalent K and K' valleys are degenerate without SOC. However, SOC lifts this degeneracy: the valence band maximum (VBM) energy is -0.123 eV at K valley and 0.0 eV at K' valley. This spontaneous breaking of valley degeneracy opens an important path for the manipulation of the valley degree of freedom.

Finally, the MoSe_2 electronic structure is shown in Figure 4 (right) as a typical representative of group VI 2H-TMDs (namely, CrS_2 , CrSe_2 , MoQ_2 , and WQ_2 , for $Q = \text{S, Se, and Te}$; see Figure S44). This materials features the well-known direct energy band gap (2.07 eV without SOC and 1.90 eV with SOC) at the degenerate K and K' valleys, which leads to the valley and spin selection through light helicity shown in Figure 5(c). An important exception of this group is CrTe_2 , for which the inclusion of SOC in HSE06 + D3 calculation closes the semiconducting band gap. A similar result was reported in a previous study⁶² with the application of SOC using full relativistic pseudopotentials. This surprising result leaves us with 14 semiconductor systems to be explored in the next sections.

3.4. Monolayer Optical Properties. The electronic structure and SOC properties manifest in the optical spectra of the semiconductor monolayers. Let us now analyze the electronic activity of the 14 TMD semiconductors (groups IV, V, and VI). The values of the optimum vacuum end dielectric constants are shown in Table S29. Figure 5 displays the colormap of the optical activity in the hexagonal BZ, obtained through eq 8, for one system representative of each group (Figures S53–S57 show the results for all of the semiconductor monolayers). The difference between absorption with σ^+ and σ^- light polarization reveals dichroism, which is largely reported around the K and K' points of the direct band gaps in group VI semiconductors.¹² This effect is a consequence of broken inversion symmetry of 2H systems together with the time reversal symmetry, and can be widely used for valleytronics applications.

Our results can be clearly divided into three groups. (a) The group IV TMDs do not present valley selection: the optical absorption at K and K' points is negligibly small, while a strong absorption is observed at M point. (b) The group V TMDs, on the other hand, feature partial valley selection. While the main absorption is observed at M points, the K and K' valleys are selectively activated by σ^- and σ^+ polarizations, respectively. (c) Finally, group VI TMDs show strong valley selection through optical helicity.

The main differences between the groups can be understood by analyzing the corresponding band structures with SOC (Figure 4). First of all, one can note that all of the systems show a characteristic valley structure with local maximum of valence bands and local minimum of conduction bands at K and K' points. The characteristic inversion symmetry breaking presented in 2H monolayers leads to non-zero but contrasting Berry curvatures and orbital magnetic moments at K/K' valleys. The SOC leads to degenerate bands in each valley, but with opposite spins. However, the valley physics show up when the band gap coincides with such valleys. The distinct number of occupied states rises the Fermi level from group IV to group VI. Therefore, in the first, the valley states are unoccupied, that is, local conduction band minimum (CBM) and VBM at K and K' points are above the Fermi level (around 2–4 eV for TiSe_2 , Figure 4); in the second, only one SOC sub-band is occupied;

in the last, both VBs are occupied and the direct band gap is observed precisely at the valleys.

3.5. Excitonic Effects. Figure 6 shows the excitonic absorption spectrum of groups IV to VI 2H-TMD monolayers,

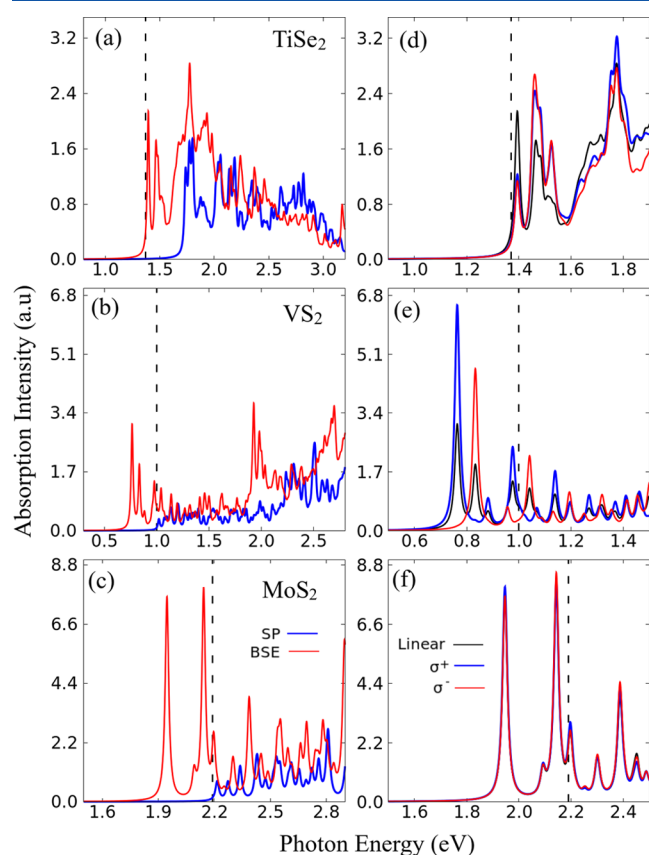


Figure 6. 2H-TMD monolayer absorption spectrum. (a–c) Comparison of exciton absorption spectrum (BSE) and a single-particle absorption spectrum for semiconductors from (a) group IV (TiSe_2), (b) group V (VS_2), and (c) group VI (MoS_2). (d–f) Effect of light polarization in excitonic absorption spectrum for the same systems.

represented by TiSe_2 , VS_2 , and MoS_2 , respectively. The spectrum of the other monolayers are similar, and can be seen in Figures S69–S73. The left panels display a comparison between the spectrum without (blue) and with (red) the excitonic effects. Both calculations are done using linear polarized light. We note that, for group IV TMDs, the first single-particle peak has higher energy than the energy gap (vertical dotted line). This is a consequence of the indirect band gap since photoexcitation preserves momentum. The TMDs of the groups V and VI, on the other hand, are semiconductors with direct band gap; therefore, the single-particle transition starts at the gap energy.

Despite the important differences between group IV, V, and VI TMD monolayers, the absorption spectra recorded with BSE (red) show strong contrast with the ones calculated for the single-particle system (blue) for all of the monolayers. This reveals the relevance of the excitons for this class of materials. Binding excitonic states with energy lower than the first single-particle photoexcitation energy manifests as a redshift of the BSE spectra, compared with the single-particle one. We can observe that the peak intensity is higher when excitonic effects are considered; this is equivalent to a higher absorption

coefficient, which shows a higher photon absorption rate for photons with the equivalent frequency.

Figure 6 (right) shows only the BSE spectra, but for different light polarization: the black curves represent linear polarization, and the red (blue) curves represent σ^- (σ^+) circular polarization. In group IV and group VI monolayers, we observe a degeneracy between linear, σ^+ , and σ^- absorption, a consequence of time reversal symmetry. In group V TMD, on the other hand, we observe that the first peaks for σ^+ and σ^- light polarization have different energies. This is a consequence of the degeneracy breaking of the energy bands near the Fermi level at the two nonequivalent valleys in VQ_2 monolayers due to ferromagnetic order.

Table 2 summarizes our main results for the excitonic properties, obtained through BSE using both PBE + D3 + SOC

Table 2. TB + BSE Calculated Excitonic Properties, with PBE + D3 + SOC + SCS and HSE06 + D3 + SOC Parameterization: Indirect (Direct) Exciton Ground State, $E_{\text{gs}}^{\text{ind(dir)}}$ (eV), and Exciton Bright Ground State, $E_{\text{gs}}^{\text{br}}$ (eV), for 2H Stable Semiconductor TMD Monolayers^a

system	XC	$E_{\text{gs}}^{\text{ind}}$	$E_{\text{gs}}^{\text{dir}}$	$E_{\text{gs}}^{\text{br}}$
TiSe_2	PBE + SCS	1.07	1.34	1.35
	HSE06	1.08	1.39	1.39
ZrTe_2	PBE + SCS	0.64	0.89	0.92
	HSE06	0.63	0.89	0.89
HfTe_2	PBE + SCS	0.48	0.92	0.94
	HSE06	0.03	0.63	0.63
VS_2	PBE + SCS	0.99	1.40	1.67
	HSE06	dir exc	0.76	0.76
VSe_2	PBE + SCS	0.23	0.67	0.71
	HSE06	dir exc	0.67	0.67
VTe_2	PBE + SCS	0.76	0.99	1.00
	HSE06	0.24	0.85	0.95
CrS_2	PBE + SCS	dir exc	1.06	1.06
	HSE06	dir exc	1.14	1.14
CrSe_2	PBE + SCS	dir exc	0.86	0.86
	HSE06	dir exc	0.79	0.79
MoS_2	PBE + SCS	dir exc	1.97	1.98
	HSE06	dir exc	1.94	1.94
MoSe_2	PBE + SCS	dir exc	1.69	1.69
	HSE06	dir exc	1.66	1.66
MoTe_2	PBE + SCS	dir exc	1.29	1.29
	HSE06	dir exc	1.21	1.21
WS_2	PBE + SCS	dir exc	2.00	2.05
	HSE06	dir exc	1.91	1.93
WSe_2	PBE + SCS	1.63	1.72	1.78
	HSE06	1.55	1.63	1.66
WTe_2	PBE + SCS	dir exc	1.19	1.26
	HSE06	dir exc	1.05	1.1

^aThe systems in bold are magnetic.

+ SCS and HSE06 + D3 + SOC. The third, fourth, and fifth columns display the lower exciton energy for indirect, direct, and bright quasi-particles, respectively. A comparison of Tables 1 and 2 reveals that indirect energy band gaps lead to indirect excitonic ground states, which means that differences in exciton binding energy do not surpass the indirect-to-direct-band gap splitting. We also note that $E_{\text{gs}}^{\text{br}}$ corresponds to the quasi-particles that satisfy the optical selection rules of conservation of linear and angular (spin plus orbital) momentum. Therefore they are the first optically active

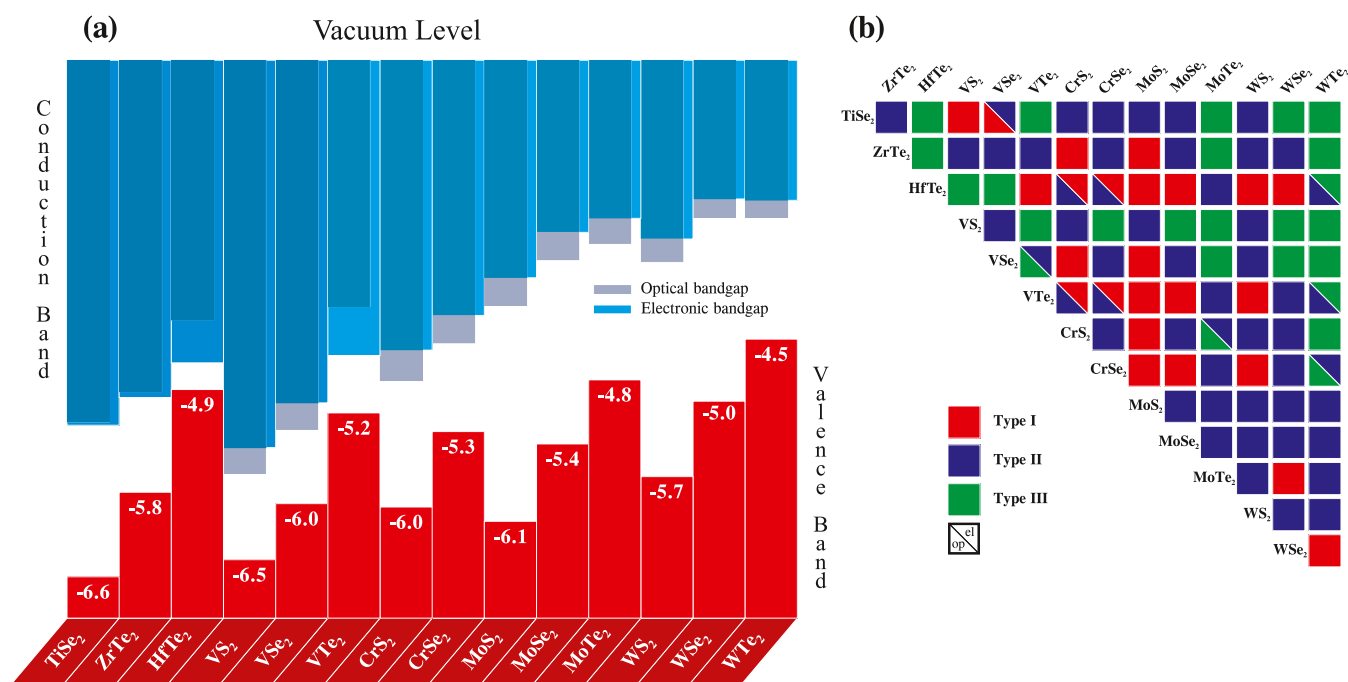


Figure 7. vdW heterostructure analysis. (a) Band offset obtained by HSE06 + D3 + SOC TB parameterization. Red bars represent VBMs, blue bars correspond to the CBMs obtained with electronic band gap, and gray bars are related to the CBMs obtained with optical band gap. The white values correspond to the ionization potential in eV. (b) Classification of semiconductor heterojunctions for combinations of the MQ₂ monolayers. The classification was obtained employing the Anderson rule and TB parameterized in DFT-HSE06 + D3 + SOC. The upper part of the squares considers the electronic band gap, and the lower part uses the optical band gap to classify the heterojunctions.

excitons; their energy coincides with the first peak at the excitonic absorption spectra (Figure 6). When $E_{\text{gs}}^{\text{dir}}$ coincides with $E_{\text{gs}}^{\text{br}}$, the first direct transition is spin allowed. Otherwise, the ground state is called dark exciton, as typically observed in W-based monolayers.⁶³

Finally, comparing PBE + D3 + SOC + SCS with HSE06 + D3 + SOC, we observe that the former functional yields a good description for excitonic effects for a majority of nonmagnetic systems. However, for magnetic systems, the energy band gap correction provided by HSE06 is spin-dependent. In this case, PBE + D3 + SOC + SCS calculation does not provide an accurate description of exciton energy and should be replaced by the hybrid functional.

3.6. TMD Heterostructures for Solar Cells. Now we consider the 91 possible bilayer heterostructure combinations composed of the 14 semiconducting monolayers and search for the best candidates for PV applications. A type II band alignment is desired for excitonic solar cells (XSCs).²⁸ To obtain the band alignment, we use the vacuum level as the reference, which was obtained by the average value of the electrostatic potential at the middle of the vacuum region.^{54,55,64} The CBM and VBM of the monolayers were obtained with respect to the vacuum level. The results for the band alignment are schematized in Figure 7. Since those systems present excitons, we distinguish the electronic CBM level (gray) from the optical CBM level (blue). While the former is obtained from the band gap of the system, the latter is obtained from the bright exciton ground state. We observe that the excitonic effects can change the band alignment.

For TMDs from group IV and VTe₂ from group V, the inclusion of excitons rises the CBM level; while the opposite effect is observed for group VI, VS₂ and VSe₂ from group V. These results indicate that only for direct energy band gap

TMDs, the exciton energy level is lower than the (direct) band gap. In group IV and VTe₂, on the other hand, the (indirect) band gap has lower energy than the first optical transition. Considering the electronic band gap, 23 heterostructures are type I, 46 are type II, and 22 are type III. If we consider optical band gap, we have 20 of type I, 48 of type II, and 23 of type III. These changes occur when the exciton binding energy is higher than the energy difference between direct and indirect band gaps and due to significant shifts in the CBM position from electronic to optical band gap.

In XSCs, the main photocurrent generation mechanism is based on the dissociation of the electron-hole pair (bounded by Coulomb interaction) into free carriers, which are dissociated into the interface, composed of the electron transport layer (ETL) and hole transport layers (HTL), on the other hand, in traditional solar cells, both electrons and holes coexist in the same chemical phase and the entire process occurs primarily in the bulk material.⁶⁵

Despite the different mechanism of photoconversion, the maximum theoretically obtainable efficiency for XSCs is given by the Shockley–Queisser limit, as it is for traditional solar cells. The only additional consideration we have taken to treat both with this approach is the substitution of electronic band gap in traditional solar cells by the optical band gap in XSCs.⁶⁵ There are different ways to estimate the PCE of an XSC.^{28,66} In this work, we use a direct approach that allows us to estimate the PCE with the data directly obtained by electronic and optical properties.

To estimate the PCE of type II TMD heterojunctions, we analyze it by considering the electronic energy band gap and the optical band gap that is obtained including excitonic effects. We follow the approach of previous works^{55,64} and

Table 3. Excitonic Effect in Type II Heterojunctions with Calculated Power Conversion Efficiency (PCE) Greater Than 10%: Donor/Acceptor Monolayers (D/A), Donor Band Gap, E_g^d (eV), Conduction Band Offset, ΔE_{CBM} (eV), Maximum Open-Circuit Voltage, V_{oc} (eV), Short-Circuit Current, J_{sc} (W/(eVm²)), Power Conversion Efficiency, PCE (%), Δ PCE (%), the Difference from Excitonic to Electronic PCE (%), Obtained Using the TB + BSE, HSE06 + D3 + SOC Parameterization

D/A	without excitons					with excitons					
	E_g^d	ΔE_{CBM}	V_{oc}	J_{sc}	PCE	E_g^d	ΔE_{CBM}	V_{oc}	J_{sc}	PCE	Δ PCE
VSe ₂ /TiSe ₂	0.91	0.20	0.41	521.61	13.89						
ZrTe ₂ /TiSe ₂	0.84	0.25	0.29	524.68	9.88	0.89	0.28	0.31	520.61	10.48	0.60
CrS ₂ /TiSe ₂	1.41	0.67	0.44	324.44	9.27	1.14	0.38	0.46	428.93	12.82	3.55
ZrTe ₂ /VSe ₂	0.83	0.05	0.48	524.68	16.70	0.88	0.34	0.24	520.61	8.45	−8.25
CrSe ₂ /CrS ₂	1.03	0.31	0.42	461.15	12.88	0.79	0.33	0.16	551.03	5.72	−7.16
VTe ₂ /CrSe ₂						0.95	0.34	0.31	509.77	10.26	
MoSe ₂ /MoS ₂	1.89	0.41	1.18	169.27	13.08	1.66	0.41	0.95	236.47	14.59	1.51
MoTe ₂ /MoS ₂	1.44	0.53	0.61	312.41	12.38	1.21	0.55	0.36	394.58	9.22	−3.16
WS ₂ /MoS ₂	2.14	0.35	1.49	116.79	11.30	1.93	0.39	1.24	161.88	13.04	1.74
WSe ₂ /MoS ₂	1.80	0.71	0.79	193.23	10.04	1.66	0.80	0.56	236.47	8.61	−1.43
MoTe ₂ /MoSe ₂	1.44	0.12	1.02	312.41	20.70	1.21	0.14	0.77	394.58	19.74	−0.96
MoSe ₂ /WS ₂	1.90	0.06	1.54	169.27	16.93	1.66	0.02	1.34	236.47	20.58	3.65
WSe ₂ /MoSe ₂	1.81	0.30	1.21	193.23	15.19	1.66	0.39	0.97	236.47	14.91	−0.28
WTe ₂ /MoSe ₂	1.25	0.29	0.66	375.22	16.09	1.10	0.38	0.42	442.28	12.01	−4.08
MoTe ₂ /WS ₂	1.44	0.18	0.96	312.40	19.48	1.21	0.16	0.75	394.58	19.22	−0.26
WTe ₂ /MoTe ₂	1.25	0.17	0.78	375.22	19.01	1.10	0.24	0.56	442.28	16.03	−2.98
WSe ₂ /WS ₂	1.81	0.36	1.15	193.23	14.43	1.66	0.41	0.95	236.47	14.61	0.18
WTe ₂ /WS ₂	1.25	0.35	0.60	375.22	14.62	1.10	0.40	0.40	442.28	11.43	−3.19

estimate the upper limit for PCE using the following equation²⁸

$$PCE = \frac{J_{sc} V_{oc} \beta_{FF}}{P_{solar}} \quad (12)$$

where β_{FF} = 0.65 is the fill factor,⁶⁷ while V_{oc} is the maximum open-circuit voltage, J_{sc} is the short-circuit current, and P_{solar} is the total incident solar power per area, defined by the following expressions

$$V_{oc} = E_g^d - \Delta E_{CBM} - 0.30 \text{ eV} \quad (13)$$

$$J_{sc} = \int_{E_g^d}^{\infty} \frac{P(E)}{E} dE \quad (14)$$

$$P_{solar} = \int_0^{\infty} P(E) dE \quad (15)$$

where E_g^d is the electronic band gap of the donor material, ΔE_{CBM} is the electronic conduction band offset, and 0.30 eV is an empirical factor to account for the loss due to energy conversion kinetics.⁶⁷

The J_{sc} expression assumes the limit of 100% for external quantum efficiency (EQE). Although it is reported that TMD 2D monolayers show an absorbance lower than 20%,¹⁹ this rate can be tuned to ~100% using light trapping schemes.⁸ $P(E)$ is the AM1.5 solar energy flux. To estimate the PCE with excitonic effects, we change the electronic band gap for the optical band gap of the monolayers, obtained by the exciton bright ground-state energy E_{gs}^{br} .

Table 3 shows the type II TMD heterojunctions with efficiency higher than 10%, considering the electronic band gap (without excitons) and optical band gap (with excitons). The empty values correspond to systems that do not form a type II heterojunction considering either electronic or optical band gap. We observe that the majority of the type II heterojunctions are formed by two direct-band-gap TMD

monolayers. The solar cells formed by Mo and W TMD monolayers show the higher efficiency, which explains the elevated number of works exploring the applications of those systems in PV devices. For some systems, the PCE obtained with excitonic effects is lower than the one obtained without these effects, which could explain the difference between experimental results for the efficiency and the simulated PCE considering only the band gap. We also note that heterojunctions with different chalcogenides feature a higher PCE for both calculation cases, which is directly correlated to the VBM position changed for each chalcogenide, which benefits the formation of type II heterojunctions. For the majority of the systems, we observe that J_{sc} is enhanced by the excitonic effects, due to the reduction of E_g^d by the same motive we observe a reduction of V_{oc} , the combination of those factors will result in an increase/decrease of PCE due excitonic effects.

4. PERFORMANCE OF THE EXCHANGE-CORRELATION FUNCTIONAL

To access the excitonic effects, we performed TB + BSE calculations based on the electronic band structure obtained via the HSE06 + D3 + SOC and PBE + D3 + SOC + SCS calculations. The former has an elevated computational cost, i.e., several times higher than PBE + D3 + SOC + SCS, in particular, due to the addition of the SOC effects, which is reduced substantially using a semilocal functional. Therefore, it is useful to identify in which cases the PBE + D3 + SOC + SCS approach yields good results.

A complete comparison between both parameterizations for all TMD monolayers is shown in Sections S7 and S9. We observe that PBE + D3 + SOC + SCS yields reliable results for the following systems: TiSe₂, ZrTe₂, CrS₂, CrSe₂, and all of the Mo- and W-based TMD monolayers. In group IV, the band structure of HfTe₂ shows a strong dependence on the functional. HSE06 lowers the VBM in *M* point and shifts the CBM, especially in *K* and *K'* points. This leads to a large shift

in exciton band structure (Figure S59). Even more important is the difference between PBE and HSE06 results for group VI CrTe₂ monolayer, since HSE06+SOC calculation changes the systems from a semiconductor to a metal, as described in Section 3.3. We also note that the hybrid functional is essential to the correct description of the magnetic materials, since it provides a different correction in band gap for each spin state.

Figure 8 displays two systems as representative examples; the blue curves represent PBE + D3 + SOC + SCS results, and

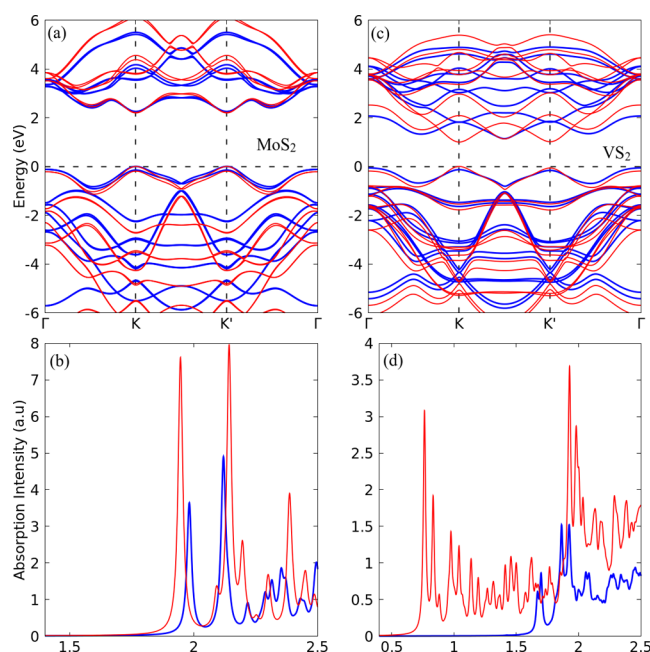


Figure 8. Comparison between TB PBE + D3 + SOC + SCS (blue curves) and HSE06 + D3 + SOC (red curves) parameterization. (a, c) Band structure of MoS₂ and VS₂ and (b, d) excitonic absorption spectrum (linear polarization) of the same systems.

the red ones correspond to the HSE06 + D3 + SOC case. In MoS₂ (left panels), PBE + D3 + SOC + SCS and HSE06 + D3 + SOC show similar band structures—especially around the Fermi level—and comparable excitonic spectra. VS₂ (right), on the other hand, display very distinct VBMs and a large redshift in the HSE06 exciton spectra.

5. INSIGHT FROM LINEAR REGRESSION ANALYSIS

In this section, we perform a linear reduction of our data to search for additional insights that they could still be able to provide in the form of correlations, as described in Section 2.4. The R^2 values for the linear regression of band gap, bright exciton ground state ($E_{\text{gs}}^{\text{br}}$), and SOC degeneracy breaking ($\delta_{k,k'}^{\text{SOC}}$) are 0.8580, 0.8806, and 0.9351, respectively. These results reveal a strong linear model that captures the general behavior of those properties, as shown in Figure 9, that compares exact values (blue) with the predictions of the linear model. Some negative values are found for $\delta_{k,k'}^{\text{SOC}}$, associated with small (>20 meV) SOC degeneracy breaking. As this model has higher R^2 and the interpolated values have no direct impact on our correlation analysis, this was considered acceptable.

From our correlation analysis, we observe that, for stable semiconductor 2H-TMDs, larger unit cells with thinner sheets have larger band gaps. The same behavior was observed for

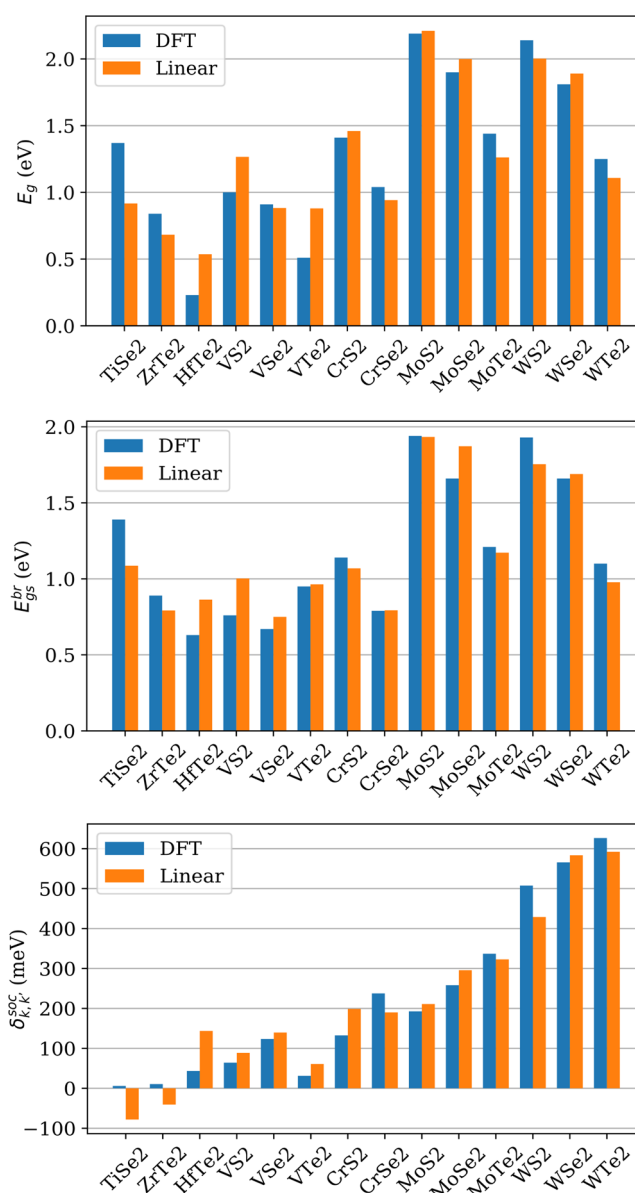


Figure 9. Visual comparison between values for the energy band gap, E_g , bright exciton ground state ($E_{\text{gs}}^{\text{br}}$), and SOC degeneracy breaking, $\delta_{k,k'}^{\text{SOC}}$, as calculated by DFT (in blue) and predicted by our linear model, eq 11 (in orange). The model aims to capture tendencies and not to reproduce each value exactly.

bright exciton ground states ($E_{\text{gs}}^{\text{br}}$), which by itself highlights a possible correlation between both in our study (in our data, we observe band gap and bright exciton ground state linearly related with $R^2 = 0.8544$). In complete opposition, we found that thicker sheets with small units cells (in area) tend to have larger SOC degeneracy breaking values. Even the two affirmations before pointing toward a possible linear dependence between SOC degeneracy breaking and band gap, our attempt to find it resulted in $R^2 = 0.3154$. This shows that even with the rule of thumb we presented here, it is possible to pinpoint some systems that have large gaps and SOC degeneracy breaking (e.g., MoSe₂ and WS₂). More details about those relations and the numerical values obtained for w_{ij} and b_i are provided in Section S10.

Focusing in the relations crossing different electronic properties, our main results come from the correlation arouse

when writing band gaps as functions of exciton energies. Throughout this work, we consider monolayers on a SiO₂ substrate. Figure 10 (green diamonds) displays the ground-

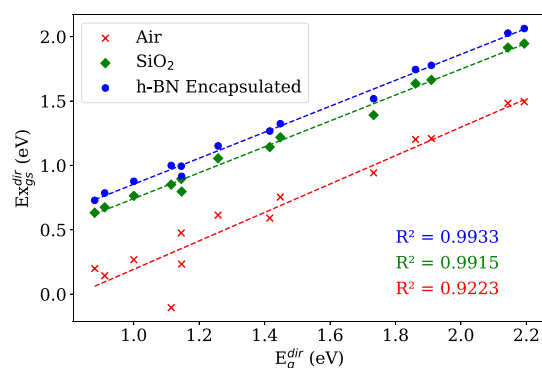


Figure 10. Direct exciton ground-state energy, Ex_{gs}^{dir} , as a function of direct band gap, E_g^{dir} , for different dielectric environment.

state energy of direct excitons, Ex_{gs}^{dir} , as a function of direct band gap, E_g^{dir} , for the 14 stable semiconductor TMDs. The direct exciton ground state may be bright (spin allowed) or dark (spin forbidden); in the former case, it corresponds to the first peak in the exciton absorption. We observe a linear behavior of Ex_{gs}^{dir} as a function of E_g^{dir} . Indeed, the exciton energy is given by $Ex_{gs}^{dir} = E_g^{dir} - E_b$, where E_b is the exciton binding energy. The linear fitting, $y = Ax - B$, of our data gives us the relation $(E_g^{dir}/\text{eV}) = (Ex_{gs}^{dir}/\text{eV}) \times 1.0055 + (-0.2637)$, with $R^2 = 0.9915$. Interestingly, our results reveal that the binding energy of the lower exciton is (264 ± 40) meV for all of the compounds we consider, irrespective of the TMD group, which indicates that the electron–hole Coulomb interaction is strongly influenced by the dielectric environment.

To confirm the effect of the substrate on the $Ex_{gs}^{dir} \times E_g^{dir}$ correlation, we repeat the calculation for the h-BN-encapsulated monolayers. The data are shown as blue circles in Figure 10. Again, we observe a good linear adjust with unitary slope $((E_g^{dir}/\text{eV}) = (Ex_{gs}^{dir}/\text{eV}) \times 1.0088 + (-0.1549))$, with $R^2 = 0.9933$. The binding energy, on the other hand, decreases to (155 ± 36) meV. Finally, to minimize the effects of the substrate, we consider the monolayer suspended in the air (red dots in Figure 10). In this case, the data show the largest deviation observed from the linear adjust $((E_g^{dir}/\text{eV}) = (Ex_{gs}^{dir}/\text{eV}) \times 1.1036 + (-0.9107))$, with $R^2 = 0.9933$, $E_b \approx (911 \pm 139)$ meV, which indicates that the binding energy features a significant dependence on the specific TMD monolayer. The dependence of the exciton absorption spectra on the dielectric environment will be further explored in a future work.

6. CONCLUSIONS

In this work, we made an extensive investigation of 2H-TMD monolayers, from group IV to group XI, including the band structure, SOC, excitonic effects, and heterostructure band alignment for photovoltaic applications. From the initial 72 monolayers considered, 22 are dynamically stable in 2H form. From those systems, an HSE06 + D3 + SOC calculation reveals that 14 (from groups IV to VI) are semiconductors, having potential for applications in solar cells.

We observed that SOC plays a minor role in the optical band gap of those systems. However, it is responsible for the valley degeneracy breaking in magnetic systems (VS₂, VSe₂, and VTe₂) and CrSe₂, which allow us to control the optical

band gap with circular light polarization. Furthermore, the absorption spectra are strongly red-shifted by excitonic effects. Both the valley degeneracy breaking driven by SOC and the excitonic correction in the optical properties may be explored to tune the power conversion efficiency in that compounds. We presented results that show good agreement with experimental data,^{68–70} for exciton energy or exciton binding energy.

For the applications in photovoltaic devices, we observed that the MoTe₂/MoSe₂, MoSe₂/WS₂, and MoTe₂/WS₂ heterojunctions show the highest efficiency in the conversion of solar energy. They feature a PCE around 20%, which suggests potential application of these materials for solar energy harvesting. Importantly, the MoSe₂/WS₂ PCE is improved by 3.65% by exciton effects. Our findings can provide guidance for future investigation and experimental development of type II TMD heterojunctions for photovoltaic applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsaem.0c03039>.

PAW projectors information; computational convergence test of the k-mesh and cutoff energy employed; lattice constants, phonon dispersions, band structure, density of states, and optical properties of TMD monolayers; and details of the simulation procedure and employed methodology (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors thank Leopoldo G. Villodre Neto for his assistance in producing Figure 7. They gratefully acknowledge support from FAPESP (São Paulo Research Foundation, Grant Numbers 2017/11631-2 and 2018/21401-7) and Shell and the strategic importance of the support given by ANP (Brazil's National Oil, Natural Gas and Biofuels Agency) through the R&D levy regulation. The authors acknowledge also the National Laboratory for Scientific Computing (LNCC/MCTI, Brazil) for providing HPC resources of the SDumont

supercomputer and Lobo Carneiro, which have contributed to the research results reported within this paper (<http://sdumont.lncc.br>). The authors thank the infrastructure provided to their computer cluster by the Department of Information Technology, Campus São Carlos.

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